Session 4 — Linear Algebra II

Exercise sheet

Work through these in order: they start with finding eigenvalues and build toward PCA and the stability of dynamical systems. The recurring idea is that eigenvalues and eigenvectors reveal the "natural directions" of a matrix — and that is what powers dimensionality reduction and tells us how neural systems evolve in time.

Exercise 1 Let $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

(a) Verify that $(1, 1)$ is an eigenvector, and give its eigenvalue.

(b) In one sentence, what does it mean for a vector to be an eigenvector of a matrix?

Exercise 2 Find both eigenvalues of $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ by solving the characteristic equation $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$.

Exercise 3 For the same matrix $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$, find an eigenvector corresponding to the eigenvalue $\lambda = 2$.

Exercise 4 Still for $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$:

(a) Compute the trace and the determinant.

(b) Verify that they equal $\lambda_1 + \lambda_2$ and $\lambda_1\lambda_2$ respectively, using the eigenvalues you found.

Exercise 5 Let $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ (symmetric).

(a) Find its eigenvalues.

(b) Find an eigenvector for each, and check that the two eigenvectors are orthogonal.

(c) Why is orthogonality guaranteed for a symmetric matrix?

Exercise 6 *A discrete linear system repeatedly applies a matrix to its state.*

The matrix has eigenvalues 0.9 (along direction $\boldsymbol{u}$) and 0.5 (along $\boldsymbol{v}$). The initial state has coefficient 1 along each eigenvector.

(a) What are the two coefficients after 1 step?

(b) After 10 steps? *(You may leave them as powers or evaluate.)*

(c) Which mode dominates the long-run behaviour, and why?

Exercise 7 *Principal Component Analysis of two correlated variables.*

The covariance matrix is $\boldsymbol{C} = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 5 \end{smallmatrix}\right)$.

(a) Find its eigenvalues.

(b) Give the direction of the first principal component.

(c) What fraction of the total variance does the first component capture?

Exercise 8 A PCA of neural data returns the eigenvalues $12,\ 6,\ 1.5,\ 0.5$ (largest first).

(a) Compute the total variance and the fraction explained by the first component.

(b) How many components are needed to capture at least 90% of the variance?

Exercise 9 A covariance matrix has eigenvalues $8,\ 5,\ 2$ (you are not told the matrix itself).

(a) What is the total variance, and which single number of the matrix equals it?

(b) Can any eigenvalue of a covariance matrix be negative? Explain.

Exercise 10 For each symmetric matrix, decide whether it *could* be a covariance matrix, based only on its eigenvalues, and say why:

(a) eigenvalues 3 and 1;

(b) eigenvalues 4 and -2;

(c) eigenvalues 2 and 0.

Exercise 11 *A recurrent network updates its activity as* $\boldsymbol{x}_{t+1} = \boldsymbol{A}\boldsymbol{x}_t$.

The connectivity matrix $\boldsymbol{A}$ has eigenvalues 0.95 and 1.05.

(a) Is the network stable (does activity settle to zero)?

(b) Describe what happens to the activity over many steps.

(c) If both eigenvalues were 0.95 instead, would it be stable?

Exercise 12 A continuous neural population obeys $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$, and $\boldsymbol{A}$ has the complex eigenvalues $-1 \pm 2i$.

(a) Is the system stable? Which part of the eigenvalue tells you?

(b) What behaviour does the imaginary part produce?

Exercise 13 A continuous system $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$ has eigenvalues 2 (along $\boldsymbol{u}$) and -3 (along $\boldsymbol{v}$). After a long time, which direction does the state align with, and does it grow or decay?

Exercise 14 *PCA on a large population recording.*

Recording 100 neurons, PCA shows that the first 3 components explain 85% of the variance.

(a) What does this say about the dimensionality of the population activity?

(b) Why is this useful for analysis and visualisation?

(c) Give one reason why choosing components by variance can be misleading.

Exercise 15 A symmetric matrix can be written as $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^\top$ with $\boldsymbol{D}$ diagonal.

(a) What do the columns of $\boldsymbol{P}$ represent?

(b) What are the diagonal entries of $\boldsymbol{D}$?

(c) Geometrically, what does applying $\boldsymbol{A}$ then do to space?